

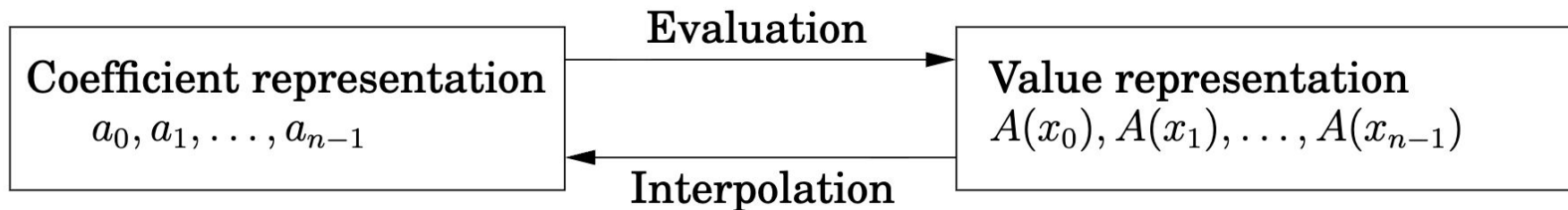


Fast Fourier Transform & Intro to Graph Algos

Lecture 4 | Algorithms Design & Analysis

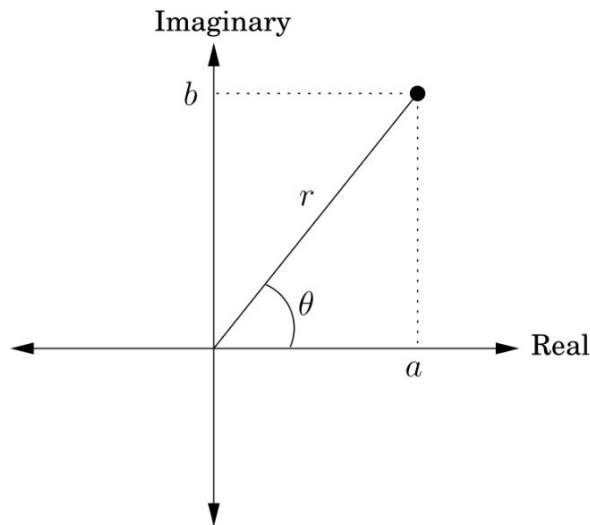
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What's Left?



<https://www.youtube.com/watch?v=h7apO7q16V0>

Complex Numbers



The complex plane

$z = a + bi$ is plotted at position (a, b) .

Polar coordinates: rewrite as $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$, denoted (r, θ) .

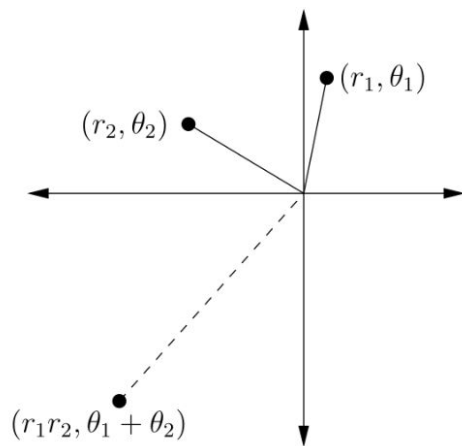
- *length* $r = \sqrt{a^2 + b^2}$.
- *angle* $\theta \in [0, 2\pi)$: $\cos \theta = a/r$, $\sin \theta = b/r$.
- θ can always be reduced modulo 2π .

Examples:

Number	-1	i	$5 + 5i$
Polar coords	$(1, \pi)$	$(1, \pi/2)$	$(5\sqrt{2}, \pi/4)$

<https://www.youtube.com/watch?v=fHAHclKdcBk>

Complex Numbers



Multiplying is easy in polar coordinates

Multiply the lengths and add the angles:

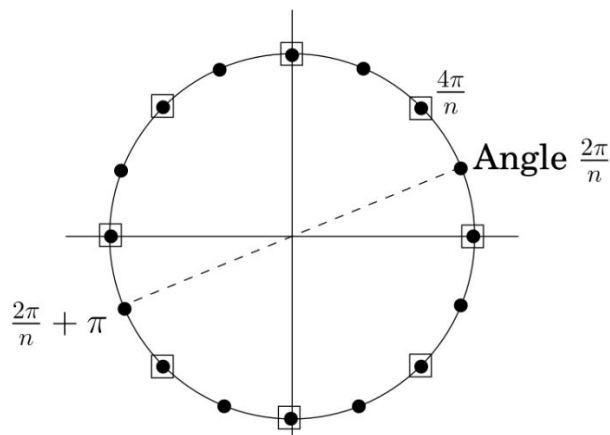
$$(r_1, \theta_1) \times (r_2, \theta_2) = (r_1 r_2, \theta_1 + \theta_2).$$

For any $z = (r, \theta)$,

- $-z = (r, \theta + \pi)$ since $-1 = (1, \pi)$.
- If z is on the *unit circle* (i.e., $r = 1$), then $z^n = (1, n\theta)$.

<https://www.youtube.com/watch?v=fHAHclKdcBk>

Roots of Unity



The n th complex roots of unity

Solutions to the equation $z^n = 1$.

By the multiplication rule: solutions are $z = (1, \theta)$, for θ a multiple of $2\pi/n$ (shown here for $n = 16$).

For even n :

- These numbers are *plus-minus paired*: $-(1, \theta) = (1, \theta + \pi)$.
- Their squares are the $(n/2)$ nd roots of unity, shown here with boxes around them.

<https://www.youtube.com/watch?v=fHAHclKdcBk>

A matrix view



$$\begin{bmatrix} A(x_0) \\ A(x_1) \\ \vdots \\ A(x_{n-1}) \end{bmatrix} = \begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n-1} & x_{n-1}^2 & \cdots & x_{n-1}^{n-1} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix}.$$

(Discrete) Fourier Transform



$$M_n(\omega) = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ & & \vdots & & \\ 1 & \omega^j & \omega^{2j} & \dots & \omega^{(n-1)j} \\ & & \vdots & & \\ 1 & \omega^{(n-1)} & \omega^{2(n-1)} & \dots & \omega^{(n-1)(n-1)} \end{bmatrix}$$

Lemma *The columns of matrix M are orthogonal to each other.*

Note: You need to use the complex conjugate definition of dot product for complex numbers.

How to find the Inverse?



The matrix can be made orthonormal by dividing by $\sqrt{n}$

Inverse of an orthonormal matrix is the Conjugate Transpose.

Inversion formula $M_n(\omega)^{-1} = \frac{1}{n} M_n(\omega^{-1}).$

Alternative view of FFT



$$\begin{array}{c}
 \begin{array}{|c|} \hline k \\ \hline \end{array} \\
 \begin{array}{|c|} \hline \omega^{jk} \\ \hline \end{array} \\
 \begin{array}{|c|} \hline j \\ \hline \end{array}
 \end{array}
 \begin{array}{|c|} \hline a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ \vdots \\ a_{n-1} \\ \hline \end{array}
 =
 \begin{array}{cc}
 \begin{array}{|c|} \hline \text{Column} \\ \hline \end{array} & \begin{array}{|c|} \hline 2k \\ \hline \end{array} & \begin{array}{|c|} \hline 2k+1 \\ \hline \end{array} \\
 \begin{array}{|c|} \hline \omega^{2jk} \\ \hline \end{array} & \begin{array}{|c|} \hline \omega^j \cdot \omega^{2jk} \\ \hline \end{array} \\
 \begin{array}{|c|} \hline j \\ \hline \end{array} & \begin{array}{|c|} \hline a_0 \\ a_2 \\ \vdots \\ a_{n-2} \\ \hline \end{array} & \begin{array}{|c|} \hline a_1 \\ a_3 \\ \vdots \\ a_{n-1} \\ \hline \end{array} \\
 \text{Even} & \text{Odd} \\
 \text{columns} & \text{columns}
 \end{array}
 =
 \begin{array}{cc}
 \begin{array}{|c|} \hline \text{Column} \\ \hline \end{array} & \begin{array}{|c|} \hline 2k \\ \hline \end{array} & \begin{array}{|c|} \hline 2k+1 \\ \hline \end{array} \\
 \begin{array}{|c|} \hline \omega^{2jk} \\ \hline \end{array} & \begin{array}{|c|} \hline \omega^j \cdot \omega^{2jk} \\ \hline \end{array} \\
 \begin{array}{|c|} \hline j \\ \hline \end{array} & \begin{array}{|c|} \hline a_0 \\ a_2 \\ \vdots \\ a_{n-2} \\ \hline \end{array} & \begin{array}{|c|} \hline a_1 \\ a_3 \\ \vdots \\ a_{n-1} \\ \hline \end{array} \\
 j + n/2 & \begin{array}{|c|} \hline \omega^{2jk} \\ \hline \end{array} & \begin{array}{|c|} \hline -\omega^j \cdot \omega^{2jk} \\ \hline \end{array} \\
 \end{array}
 \end{array}$$

Alternative view of FFT



$$\begin{array}{c}
 \text{Row } j \\
 \hline
 \begin{array}{c}
 \boxed{M_{n/2}} \begin{array}{c} a_0 \\ a_2 \\ \vdots \\ a_{n-2} \end{array} \\
 + \omega^j \boxed{M_{n/2}} \begin{array}{c} a_1 \\ a_3 \\ \vdots \\ a_{n-1} \end{array}
 \end{array} \\
 \\
 \begin{array}{c}
 j + n/2 \\
 \hline
 \begin{array}{c}
 \boxed{M_{n/2}} \begin{array}{c} a_0 \\ a_2 \\ \vdots \\ a_{n-2} \end{array} \\
 - \omega^j \boxed{M_{n/2}} \begin{array}{c} a_1 \\ a_3 \\ \vdots \\ a_{n-1} \end{array}
 \end{array}
 \end{array}$$

Alternative view of FFT



Figure 2.9 The fast Fourier transform

function FFT (a, ω)

Input: An array $a = (a_0, a_1, \dots, a_{n-1})$, for n a power of 2

A primitive n th root of unity, ω

Output: $M_n(\omega) a$

if $\omega = 1$: return a

$(s_0, s_1, \dots, s_{n/2-1}) = \text{FFT}((a_0, a_2, \dots, a_{n-2}), \omega^2)$

$(s'_0, s'_1, \dots, s'_{n/2-1}) = \text{FFT}((a_1, a_3, \dots, a_{n-1}), \omega^2)$

for $j = 0$ to $n/2 - 1$:

$r_j = s_j + \omega^j s'_j$

$r_{j+n/2} = s_j - \omega^j s'_j$

return $(r_0, r_1, \dots, r_{n-1})$

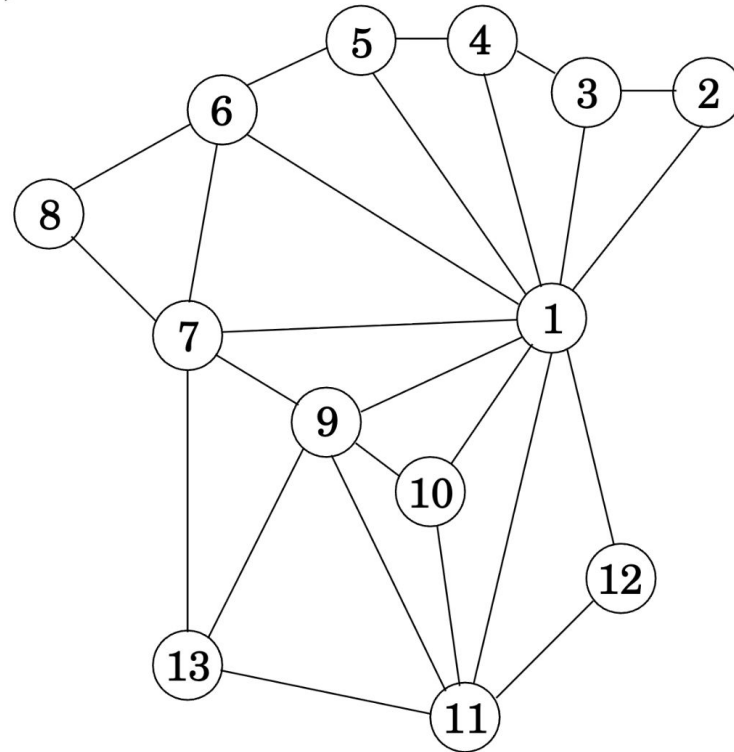


Graphs

Graph Abstraction



(b)



Graph Representation



We can represent a graph by an *adjacency matrix*; if there are $n = |V|$ vertices $v_1, \dots, v_n$, this is an $n \times n$ array whose (i, j) th entry is

$$a_{ij} = \begin{cases} 1 & \text{if there is an edge from } v_i \text{ to } v_j \\ 0 & \text{otherwise.} \end{cases}$$

For undirected graphs, the matrix is symmetric since an edge $\{u, v\}$ can be taken in either direction.

The biggest convenience of this format is that the presence of a particular edge can be checked in constant time, with just one memory access. On the other hand the matrix takes up $O(n^2)$ space, which is wasteful if the graph does not have very many edges.

An alternative representation, with size proportional to the number of edges, is the *adjacency list*. It consists of $|V|$ linked lists, one per vertex. The linked list for vertex u holds the names of vertices to which u has an outgoing edge—that is, vertices v for which $(u, v) \in E$.

Explore Graphs



procedure explore(G, v)

Input: $G = (V, E)$ is a graph; $v \in V$

Output: $\text{visited}(u)$ is set to true for all nodes u reachable from v

$\text{visited}(v) = \text{true}$

$\text{previsit}(v)$

for each edge $(v, u) \in E$:

 if not $\text{visited}(u)$: $\text{explore}(u)$

$\text{postvisit}(v)$

procedure previsit(v)

$\text{pre}[v] = \text{clock}$

$\text{clock} = \text{clock} + 1$

procedure postvisit(v)

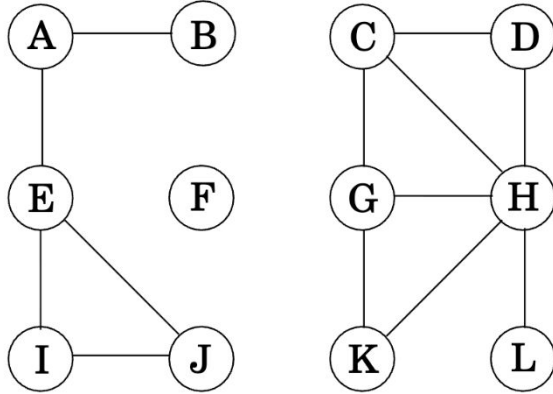
$\text{post}[v] = \text{clock}$

$\text{clock} = \text{clock} + 1$

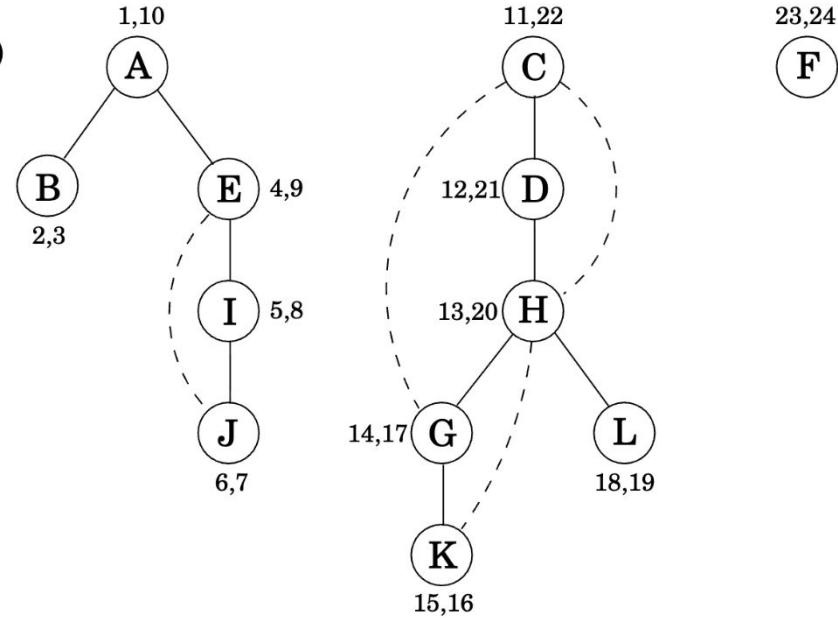
DFS Forest



(a)



(b)

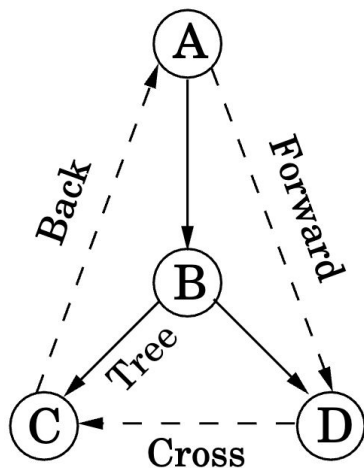


Property For any nodes u and v , the two intervals $[pre(u), post(u)]$ and $[pre(v), post(v)]$ are either disjoint or one is contained within the other.

DFS Tree Edge types



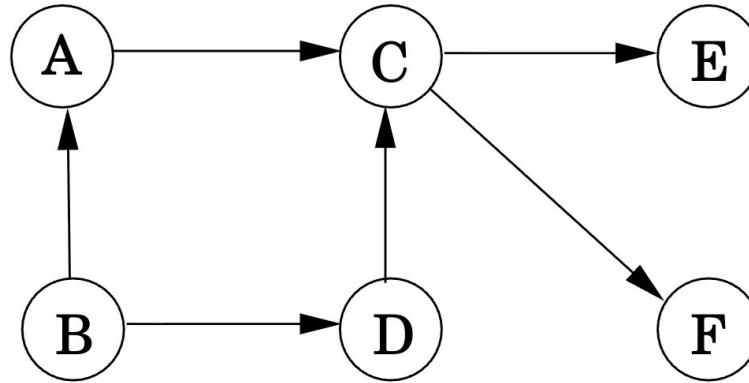
DFS tree



pre/post ordering for (u, v)				Edge type
$\left[\begin{array}{c} \\ u \end{array} \right]$	$\left[\begin{array}{c} \\ v \end{array} \right]$	$\left] \begin{array}{c} \\ v \end{array} \right]$	$\left] \begin{array}{c} \\ u \end{array} \right]$	Tree/forward
$\left[\begin{array}{c} \\ v \end{array} \right]$	$\left[\begin{array}{c} \\ u \end{array} \right]$	$\left] \begin{array}{c} \\ u \end{array} \right]$	$\left] \begin{array}{c} \\ v \end{array} \right]$	Back
$\left[\begin{array}{c} \\ v \end{array} \right]$	$\left] \begin{array}{c} \\ v \end{array} \right]$	$\left[\begin{array}{c} \\ u \end{array} \right]$	$\left] \begin{array}{c} \\ u \end{array} \right]$	Cross

Property A directed graph has a cycle if and only if its depth-first search reveals a back edge.

Directed Acyclic Graphs (DAGs)



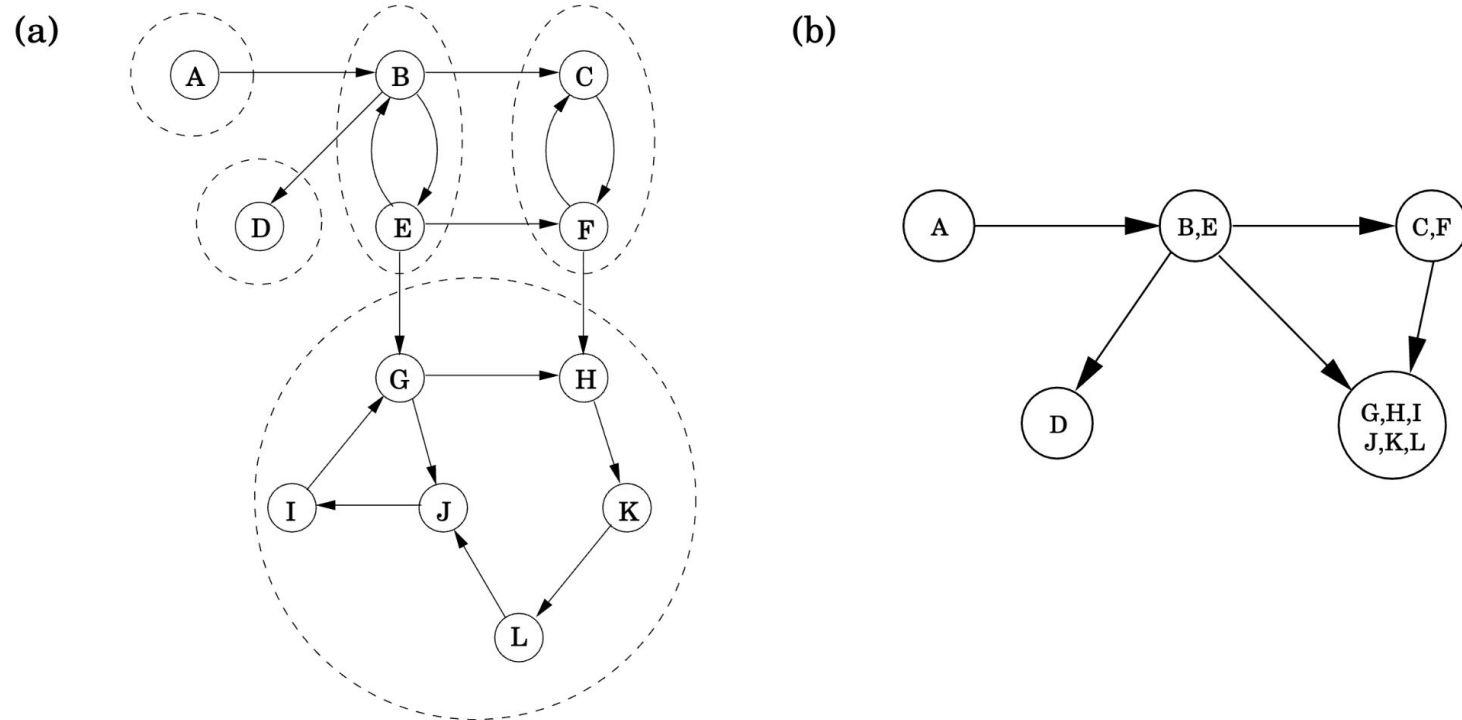
Property *A directed graph has a cycle if and only if its depth-first search reveals a back edge.*

Property *In a dag, every edge leads to a vertex with a lower post number.*

Strongly Connected Components



Figure 3.9 (a) A directed graph and its strongly connected components. (b) The meta-graph.



Backtracking

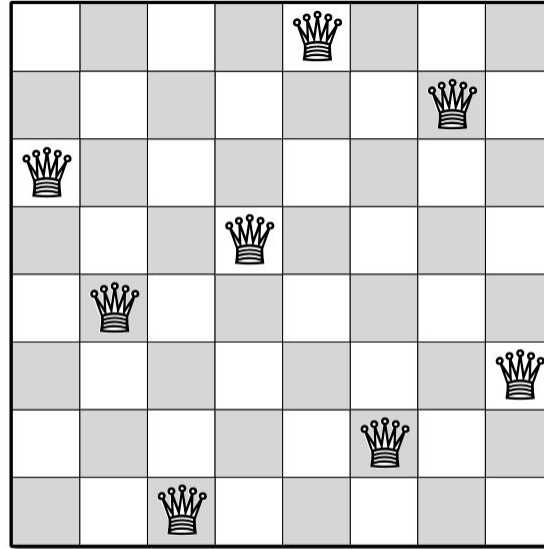


Figure 2.1. Gauss's first solution to the 8 queens problem, represented by the array $[5, 7, 1, 4, 2, 8, 6, 3]$

Backtracking



PLACEQUEENS($Q[1..n], r$):

if $r = n + 1$

 print $Q[1..n]$

else

 for $j \leftarrow 1$ to n

$legal \leftarrow \text{TRUE}$

 for $i \leftarrow 1$ to $r - 1$

 if $(Q[i] = j) \text{ or } (Q[i] = j + r - i) \text{ or } (Q[i] = j - r + i)$

$legal \leftarrow \text{FALSE}$

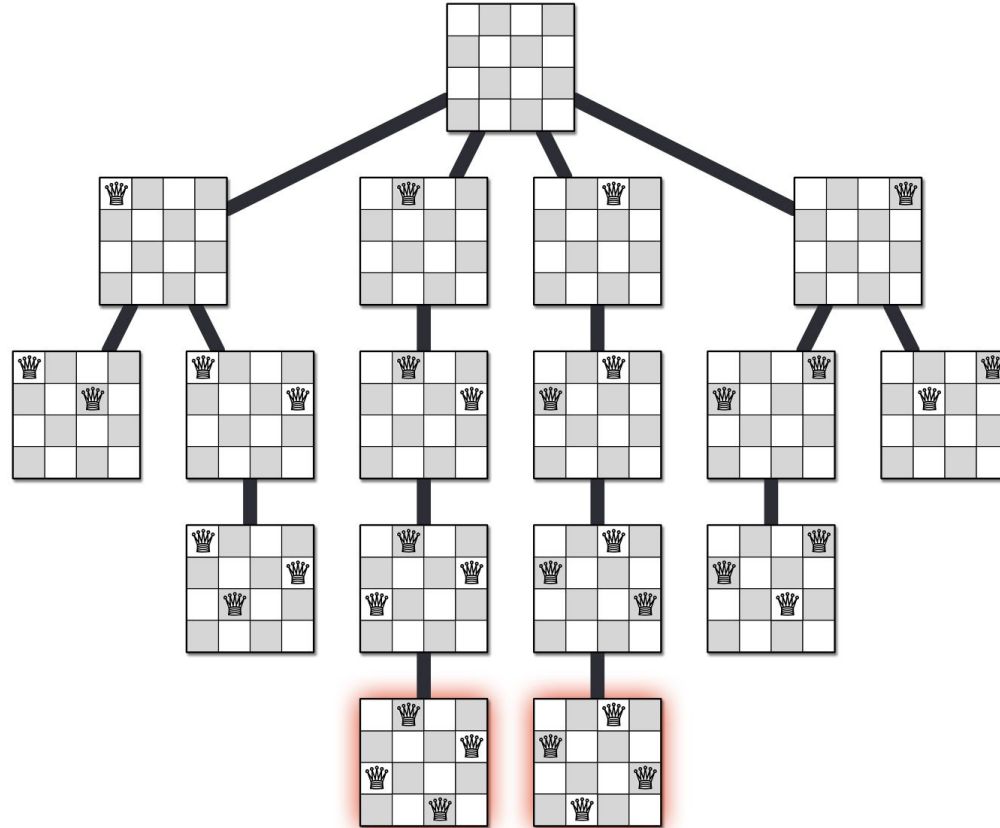
 if $legal$

$Q[r] \leftarrow j$

 PLACEQUEENS($Q[1..n], r + 1$) *⟨⟨Recursion!⟩⟩*

Figure 2.2. Gauss and Laquière's backtracking algorithm for the n queens problem.

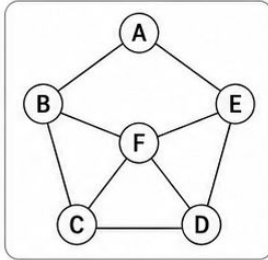
Recursion Tree



3-COLORING USING BACKTRACKING

1 Given graph

We want to assign one of the 3 colors to each vertex so that no two adjacent vertices have the same color.



Color 1

Color 2

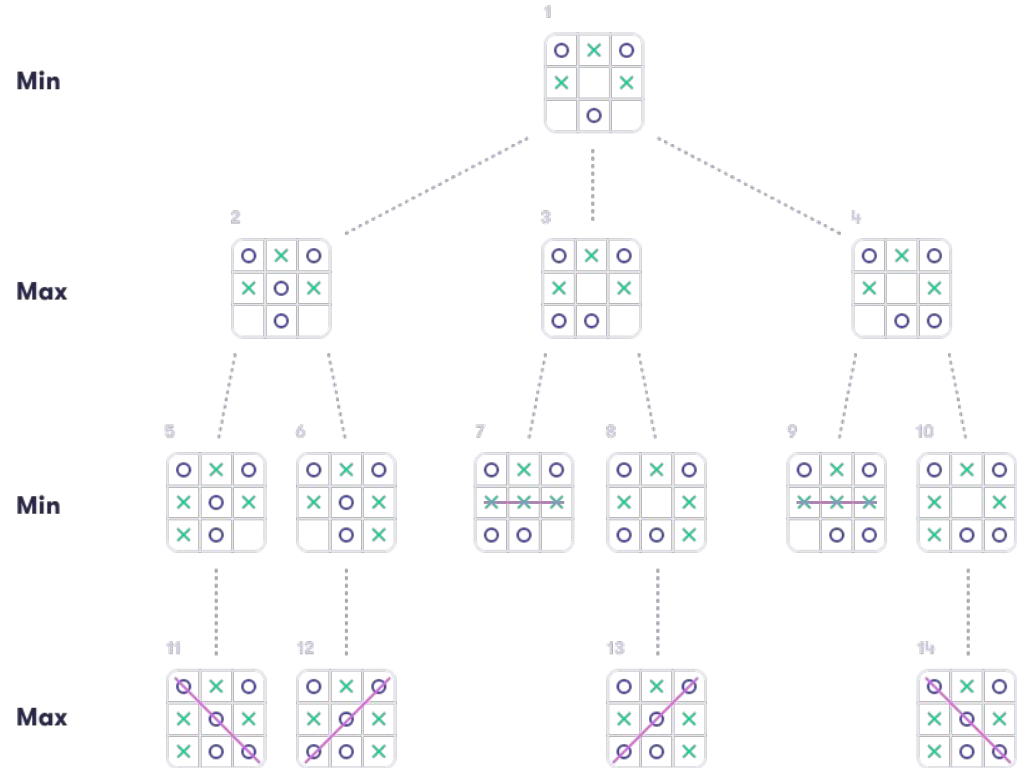
Color 3

Uncolored

Game Trees



<https://course.elementsofai.com/2/3>



Read



Chapter 2, Section 2.6, Chapter 3 in

<http://algorithmics.isi.upc.edu/docs/Dasgupta-Papadimitriou-Vazirani.pdf>

Solve



Exercises 3.3, 3.6, 3.7, 3.8, 3.16, 3.26, 3.28, in

<http://algorithmics.isi.upc.edu/docs/Dasgupta-Papadimitriou-Vazirani.pdf>